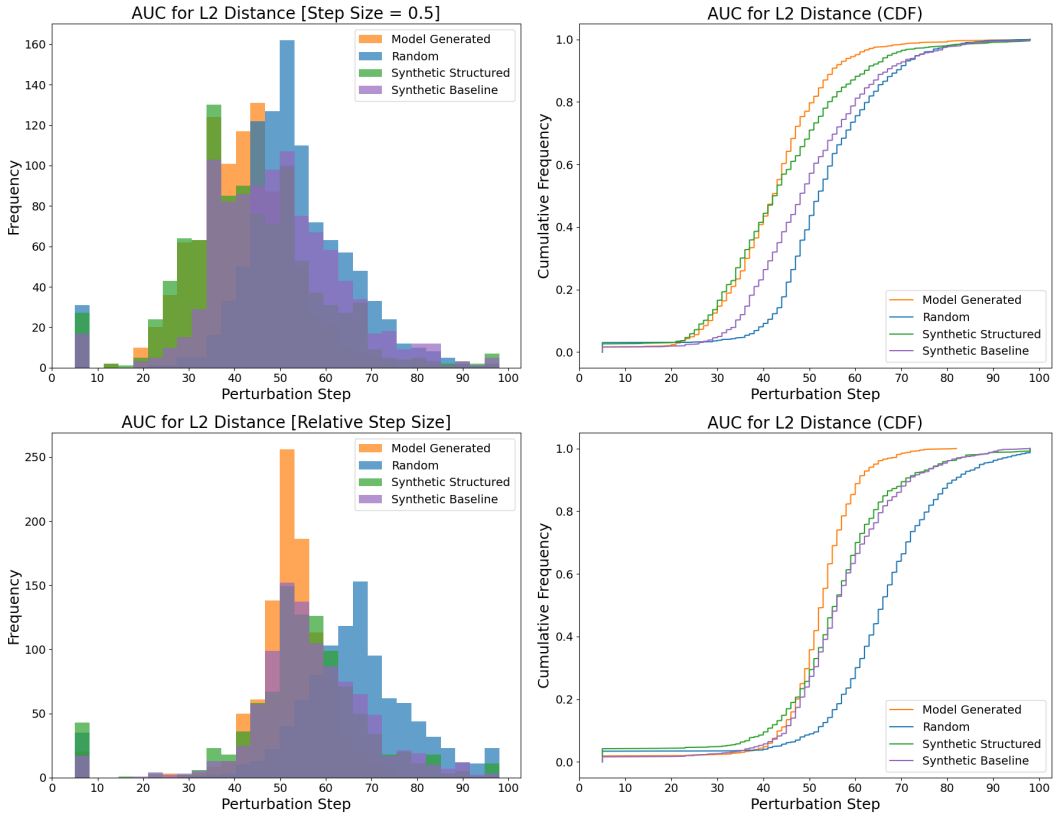

Figure B.1: The distributions of the AUC steps for perturbations with absolute step size (top) and relative step size (bottom) towards model-generated (orange), random (blue), synthetic-baseline (purple), and synthetic-structured (green) activations. The left column shows the counts of AUC steps occurring in different bins along the length of the perturbation, and the right column shows the cumulative frequency for the same. We find that our results for the AUC step distributions are similar to those for the MS step distributions.

B.2 Non-Linear (NL)

Using L2 distance to observe the perturbations reveals that the region before the blowup is not flat, but linear with varying slopes (Figure 1). In order to study the size of the initial linear portion of the curves, we use the Non-Linear (NL) metric, which points to the earliest step at which the slope of the L2 distance vs perturbation step curve deviates from linearity by more than 10% of the initial slope. We use this metric as an alternate measure for the size of the activation plateau around the base activation along different perturbation directions.

We observe that perturbations towards model-generated activations cause the quickest deviation from linearity followed by synthetic-structured activations, which is in line with our previous results for blowup locations (Figure B.2, Table B.2). However, we find that the deviation from linearity occurs the latest during perturbations towards synthetic-baseline activations, which suggests that L2 distance has a higher initial slope for these perturbations, giving more room for changes in the slope before they are classified as a deviation from linearity. In this case, the behavior of synthetic-baseline activations provides further evidence that local relationships between SAE latents are important to approximate model-generated activations.

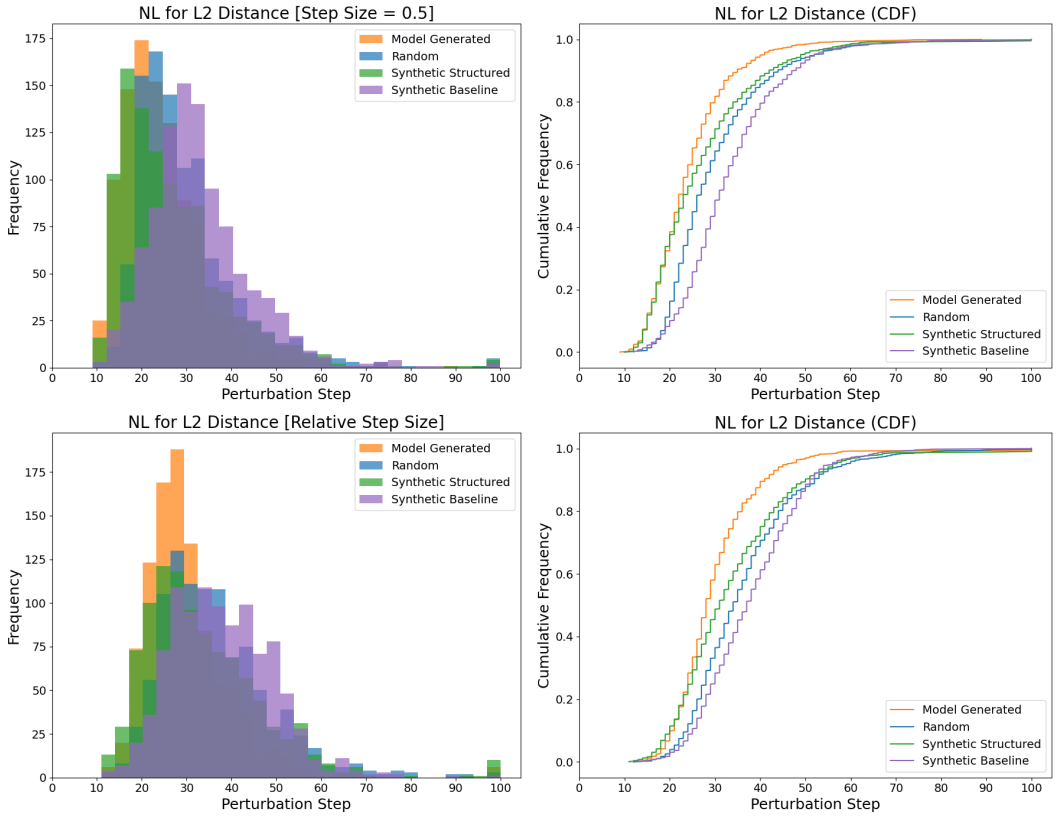

Figure B.2: The distributions of the NL steps for perturbations with absolute step size (top) and relative step size (bottom) towards model-generated (orange), random (blue), synthetic-baseline (purple), and synthetic-structured (green) activations. The left column shows the counts of NL steps occurring in different bins along the length of the perturbation, and the right column shows the cumulative frequency for the same. We find that synthetic-structured activations and random activations behave more like model-generated activations than synthetic-baseline activations do.

Non-Linear (NL) Step Distribution Statistics

Activation Type	Absolute Step Size			Relative Step Size		
	Mean	Std dev	KS	Mean	Std dev	KS
Model Generated	24.17	8.87	0.00	29.98	9.95	0.00
Random	29.80	11.72	0.22	36.33	12.33	0.27
Synthetic Baseline	32.90	11.25	0.40	37.91	10.96	0.37
Synthetic Structured	26.72	12.25	0.11	33.69	13.44	0.17

Table B.2: In terms of NL step distributions, we find that synthetic-structured activations perform better than random activations, but synthetic-baseline activations do not. This table contains the mean, standard deviation and KS statistic for NL step distributions for all the perturbations we perform. The KS statistic is measured against perturbations towards model-generated activations, with a lower value meaning higher similarity.

C KL Divergence

While previous works have predominantly used KL divergence as a measure of sensitivity, our analysis revealed potential limitations of this approach. We observed that KL divergence produces a step-function-like curve even when linear perturbations are performed at the final layer of the model right before the unembedding. This behavior suggests that the step-function shape might be an artifact of the KL divergence metric itself (or possibly due to softmax), rather than a true representation

of activation plateaus. The logarithmic nature of KL divergence may amplify differences as they become larger, leading to a more pronounced blowup region and a flatter initial plateau region.

With the mentioned caveats in mind, we perform perturbations at Layer 1 and observe their effect on KL divergence of the logits distribution instead of L2 distance at Layer 11. Figure C.1 illustrates the MS step distribution for KL divergence across different activation types. KL divergence blowups are more localized in the relative step size setup than L2 distance blowups, suggesting that the model's output distribution is more robust to noise than the model's final layer activations, only blowing up when more than 40% of the base activation has been replaced. Similar to the results for L2 distance, we find that perturbations towards synthetic-structured activations are more similar to perturbations towards model-generated activations than synthetic-baseline activations are. The difference between synthetic-structured and synthetic-baseline activations is more pronounced for KL divergence than L2 distance.

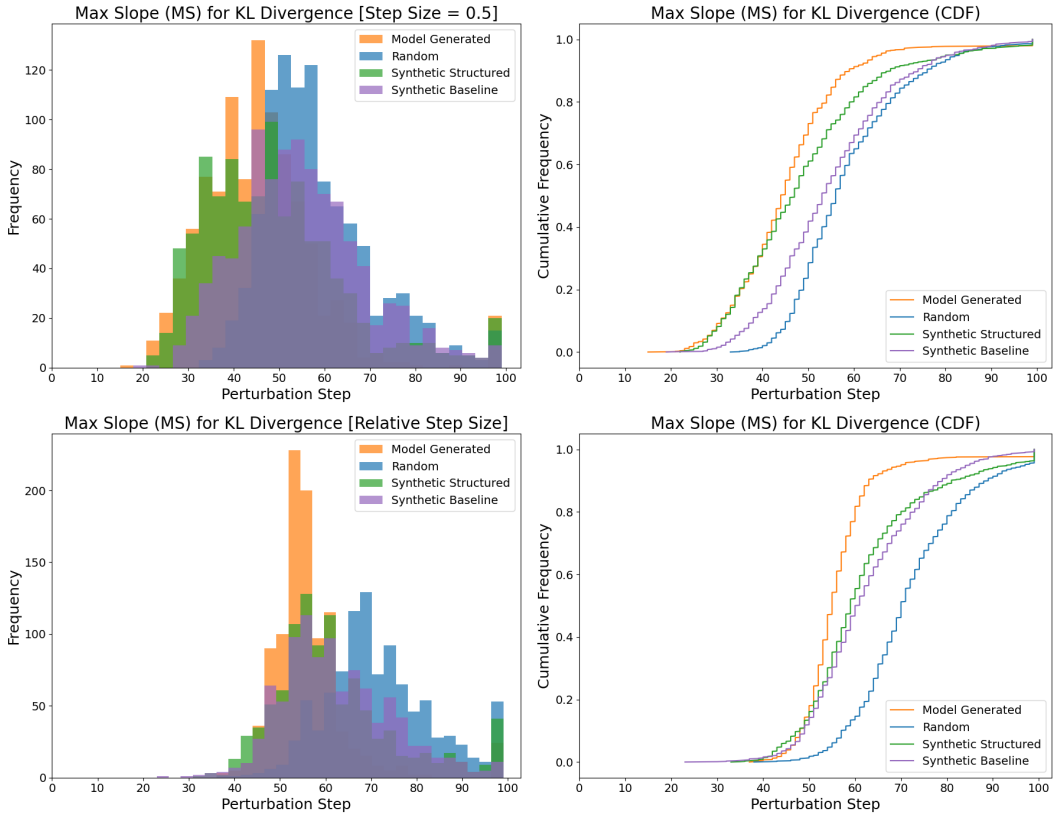

Figure C.1: The distributions of the MS steps for KL divergence of next-token prediction probabilities for perturbations with absolute step size (top) and relative step size (bottom) towards model-generated (orange), random (blue), synthetic-baseline (purple), and synthetic-structured (green) activations. The left column shows the counts of MS steps occurring in different bins along the length of the perturbation, and the right column shows the cumulative frequency for the same. We find that our results for KL divergence are similar to those for L2 distance.